

# Plan: Gömb tuner — parameter count visibility and SNAP joint AUC search

**Date:** 2026-05-12    **Slug:** gomb-tune-n-params-auc

## Scope and goal

- Surface `n_params` from each `run_gomb_smoke` JSON row in `run_gomb_tune` trial logs and per-dataset phase summaries.
- Mildly expand compact search on Slashdot/Epinions (extra `d_embed` choice; joint compact `topk` cap 24  $\rightarrow$  28) to seek higher test AUROC under existing joint-slot subsample contract.
- Bump default Slashdot trial/epoch counts in `run_gomb_external_auc_tuning.sh` for deeper random search.

## Affected files

`signedkan.wip/src/run_gomb_tune.py`, `signedkan.wip/experiments/run_gomb_external_auc_tuning.sh`, `signedkan.wip/tests/test_hymeko_gomb.py`, `docs/plans/2026-05-12-gomb-tune-n-params-auc/*`, `reports/2026-05-12-gomb-tune-n-params-auc.md`.

## CORE.YAML

None touched.

## Interface changes

None to public Rust; Python-only: internal helper `_n_params_from_row`; behaviour change: SNAP joint compact `topk` clamp constant; optional new summary keys `best_n_params`.

## Test strategy

Unit tests: `_n_params_from_row` parsing; update joint compact clamp assertion ( $\leq 28$ ); regression: existing tuner/smoke tests pass.

## Performance budget

Subprocess RSS unchanged in intent; optional `topk` up to 28 may increase VRAM vs 24 — user requested AUC tuning; cap remains below full-menu 40.

## Rollback path

Revert clamp constant and `d_embed` menu; remove log/summary fields.

## Risks

Joint compact `topk=28` may OOM on smallest GPUs; mitigated by `joint_slot_cap` and walk caps unchanged.